TT–I–544

FEBRUARY 23, 1956

SUPERSEDING
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August 30, 1954

FEDERAL SPECIFICATION

INK; MARKER, FELT TIP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers quick-drying marking ink for use with felt-tip fountain-type markers on porous and nonporous surfaces.

1.2 Classification.

1.2.1 *Type.* — The marking ink shall be of but one type.

1.2.2 *Colors.*—(See 3.3.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. — The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG–M–117—Marker, Fountain Type, Felt Tip.

CCC–T–191—Textile Test Methods.

PPP–C–96—Cans. Metal, 28 Gage and Lighter.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Material. — The manufacturer is given latitude in the selection of raw materials and processes of manufacture, provided the product meets the requirements of this specification.

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3.2 Composition.—The ink shall consist essentially of a dye dissolved in an organic solvent. Binding ingredients to provide surface adhesion may be added.

3.3 Colors. — Unless otherwise specified, marking ink shall be furnished in one of the following colors, as specified in the invitation for bids: Black, blue, green, red, and purple. (See 6.2.)

3.3.1 *Markings.* — Markings made as described in 4.4.1 shall be of the color specified and clearly legible.

3.4 Toxicity.—The ink shall contain no benzol (benzene), aniline oil, chlorinated compounds, or other toxic or hydrolyzable chlorine derivatives. The suppliers shall furnish to the contracting officer a certificate of compliance with the requirements of this paragraph.

3.5 Drying time. — Ink on panels prepared and tested in accordance with 4.4.2 shall not smear or smudge the writing paper used in the test.

3.6 Resistance to water. — Ink on panels prepared and tested in accordance with 4.4.3 shall not smear or smudge the writing paper used in the test. A slight dulling in color of the ink on the panels is permissible.

3.7 Light fastness.—The ink shall show no evidence of a marked color change when tested as described in 4.4.4. Slight changes in the brightness are permissible, but the hue shall remain constant.

3.8 Clogging of markers. — Markers shall show no evidence of clogging or difficulty in starting ink flow after being unused for a period of 96 hours. (See 4.4.5.)

3.9 Effect on applicator.—The ink shall not contain ingredients which have a deleterious effect on fountain markers used in its application. (See 4.4.5.)

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of all ink of the same color manufactured as one batch and offered for delivery at one time. If the ink cannot be identified by batch, a lot shall consist of not more than 100 gallons of each color of ink offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken from each lot by the Government inspector in accordance with Military Standard MIL–STD–105 at Inspection Level I and Acceptable Quality Level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking, and other requirements not involving tests.

4.1.3 *Sampling for tests.*—From each inspection lot, the Government inspector shall take two containers at random. From each of the two containers, one-pint specimens shall be taken and placed in separate clean, dry, metal or glass containers, sealed, marked, and forwarded to the testing laboratory designated by the bureau or agency concerned. If less than one-pint containers are used, enough containers to form two separate pints shall be taken.

4.2 Inspection. — Each sample filled container, taken as specified in 4.1.2, shall be examined for defects of construction of the container and the closure for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects, or under required fill, shall be rejected, and if the number of rejected containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for inspection; provided the contractor has removed (replaced)

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all nonconforming products. Other packaging and packing requirements specified in section 5 shall also be determined by inspection.

4.3 Lot acceptance tests.—The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4. If either specimen fails one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.4 Tests.

4.4.1 *Preparation of test panels.*—A clean fountain-type marker shall be filled with the ink to be tested. The marker shall conform to type I of Federal Specification GG–M–117. Markings shall be made on water-resistant solid-fiberboard panels, veneer-plywood, burlap, and on cleaned tin-plate panels.

4.4.2 *Drying time.*—Prepare panels as specified in 4.4.1 and allow all panels except tin-plate panel to dry for 1 minute at a room temperature of 70° to 90°F. Allow tin-plate panel to dry 2 minutes. Immediately after the specified drying time, place a sheet of white chemical wood writing paper over the markings. Place a 100-gram weight on one edge of the paper and draw the sheet of paper and the weight slowly across the markings. Examine the paper for smudge.

4.4.3 *Resistance to water.*—Panels prepared as specified in 4.4.1 shall be allowed to dry at a room temperature of 70° to 90°F. for one hour. Panels shall then be immersed to a depth corresponding to two-thirds of the height of the markings for 4 hours. Remove panels at end of immersion period and dry at a temperature of 70° to 90°F. for one hour. Test for smudging as described in 4.4.2 and examine the panels.

4.4.4 *Light fastness.*—Solid-fiberboard panel prepared as specified in 4.4.1 shall be allowed to dry at a room temperature of 70° to 90°F. for one hour. Two test specimens, 2½ by 3 inches, with markings, shall be cut from the panel. One specimen shall be exposed for 96 hours to light as described in method 5660 of Federal Specification CCC–T–191 and the other specimen retained but not exposed. The two specimens shall then be examined to determine conformance with 3.7.

4.4.5 *Clogging and deterioration of markers.*—Fill clean markers, conforming to type I, and type III, style B, of Federal Specification GG–M–117, with ink to be tested and make sufficient markings to thoroughly saturate the felt nib. Replace covers on markers and allow to stand undisturbed with the felt tip up for 96 hours. After this period, remove cover and make new markings. A few initial strokes shall be made to permit flow of ink to saturate nib. The felt nib shall be examined for breakage and evidence of disintegration after the exposure period and during the preparation of markings. Clean the markers thoroughly and examine for pitting and deterioration of the nib.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *For domestic shipment.*—Unless otherwise specified, marking ink shall be furnished in 1-pint, 1-quart, or 1-gallon containers in accordance with the manufacturer's commercial practice, as specified in the invitation for bids.

5.1.2 *For domestic shipment and storage, and overseas shipment.*—Marking ink shall be packaged in 1-pint, 1-quart, or 1-gallon metal containers conforming to type V of Federal Specification PPP–C–96, exterior coating to be plan A or B as applicable; size of container to be specified in the invitation for bids.

5.2 Packing.

5.2.1 *For domestic shipment (immediate use).*—Marking ink, packaged as specified in 5.1.1, shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Consolidated Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

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5.2.2 *For domestic shipment and storage, and overseas shipment.*—Marking ink, packaged as specified in 5.1.2, shall be packed in accordance with the applicable requirements of Appendix I of Federal Specification PPP-C-96.

5.3 Marking.

5.3.1 *Unit containers.*—Unless otherwise specified, unit containers shall be marked with the name of the material, color, quantity contained therein, month and year of manufacture, and the name of the contractor or manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name and color of the material as defined by the contract or order under which the shipment is made, quantity contained therein, date, name of the contractor, number of the contract or order, name and address of the consignee, and the gross weight.

5.3.2.1 *Additional marking (Army, Navy, and Air Force).*—In addition to the requirements in 5.3.2, and any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129. Nomenclature shall be indicated as "INK; MARKER, FELT TIP".

6. NOTES

6.1 Intended use.—Marking ink covered by this specification is intended for use in felt-tip marking pens for freehand marking, writing, or drawing on materials such as fiberboard, paper, wood, burlap, or metal.

6.2 Ordering data.—Purchasers should specify color of ink required, size of container, and should exercise any desired options offered herein. (See 1.2, 3.3, 5.1, 5.2 and 5.3.)

6.3 Transportation description.—Transportation description applicable to this item is:

Ink, marking.
Carload minimum weight 36,000 pounds.
Truckload minimum weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—Q M
Navy—S MC Or Sh
Air Force.

Copies of this specification may be purchased for 10 cents.

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